

Self-critical rumination - A network approach

2. Methods

Psychological networks have their own characteristics which differentiate them from other kinds of networks of graph theory (Epskamp et al. (2017a); Epskamp & Fried (2018); Newman (2010)). Vertices (also called nodes) usually represent some psychological measures or items of a psychological assessment tool, while the edges of the network reflect the corresponding estimates of association derived from the data. Having two rumination-related questionnaires, the **Ruminative Response Style** (*RRS*) and the **Self-critical Rumination Scale** (*SCRS*), we assumed the following hypotheses:

- Network analysis will reveal their respective internal structure previously established in the literature (such as *RRS* having a two-scale structure: *reflection* and *brooding*; while *SCRS* will show a single-scale structure);
- on the other hand, we hypothesized that in their common network, items overlapping in scope will change these clear outlines of the two measurement tools. According to our hypothesis, items of the *RRS* scale *brooding* will have non-trivial association with the self-critical aspects of rumination;
- we also assumed a combined rumination network, representing the generic (more trait-like) and the self-critical aspects of rumination will exhibit a three-component structure.

In the current study, we used a fairly large sample of individuals (N=851) to construct a dense, undirected network of rumination-related psychological constructs. The vertices represent the items of the *RRS* and *SCRS*, while edges are penalized partial correlations (using LASSO penalization, see Friedman et al. (2007); Tibshirani (1996)).

2.1. Building networks with the Least Absolute Shrinkage and Selection Operator (LASSO)

We constructed individual and combined networks of *RRS* and *SCRS* with respective penalized correlation matrices using the *EBICglasso* function of the **qgraph** package (Epskamp et al. (2012); see our Git repository for other parameter specifications). In order to shrink the edges, achieve high specificity, and get a less dense network, we let the function to exclude any elements of the underlying precision matrix below a threshold value (*threshold* parameter set to TRUE).

The **LASSO** uses *L1* (absolute) regularization (Tibshirani (1996), Friedman et al. (2007)). In the current study, the method resulted in a partial correlation network where edges represent direct associations between pairs of items, while the influence of all other items in the network are kept controlled. The regularization by *L1* penalty lets trivially small partial correlations to be set to zero (McNally et al. (2017)), thus ensure a *sparse* network which are more interpretable (Epskamp & Fried (2018)).

The tuning parameter of *gamma* in the graphical LASSO algorithm has been set to 0.5 value in accordance with (McNally et al. (2017) Bernstein et al. (2019)). The final networks were chosen by the lowest values of the *extended Bayesian Information Criterion* (EBIC).

2.2 Redundant items

We applied the *goldbricker* function from the R package *networktools* (Jones, 2017) in order to search for pairs of items in the *RRS*, *SCRS* and their combined networks that were highly correlated ($r > 0.75$) and exhibited the same pattern with other items.

After the LASSO regularization no further dimensionality reduction was necessary based upon the *goldbricker* evaluation.

2.3 Network descriptives and community detection

A comparative overview of the three networks was produced with their descriptives (degree distribution) and their centrality measures (strength, closeness, betweenness; Kolaczyk (2020)), and their One- and Two-Step Expected Influence (EI) measures to understand the cumulative importance of vertices (Donald J. Robinaugh (2016), Heeren et al. (2018), Bernstein et al. (2019)). In order to further elaborate the question of vertex importance, we identified *bridge nodes* using the functions provided by the *networktools* R package (Jones (2017)) to identify bridge vertices (connectors between sub-graphs, see an example at Bernstein et al. (2019)). We estimated both one-step and two-step bridge expected influence measures which reflect the sum of edge weights connecting one vertex to other groups on vertices (or communities); and also express a vertex's secondary influence on the rest of the network, respectively. Higher influence scores indicate that a certain vertex is more likely to active neighbouring vertices or communities. In order to estimate the Expected Influence metrics, we used the *expectedInf* function of the *networktools* R package. In order to get an understanding of community of vertices and to see if vertices can be grouped into neighbourhoods corresponding to the previously established sub-scale structure of *RRS*, *SCRS* and investigate their separation in their combined network, we used the community detection methods available in the *igraph* R package (Kolaczyk (2020)). When detecting possible communities of vertices, we followed guidelines by Reichardt & Bornholdt (2006), Bernstein et al. (2019) and Kolaczyk (2020). In order to mitigate the effect of choice of community detection algorithms, we performed more of them available in the *igraph* package multiple times.

2.4 Internal Structure with an exponential random graph model

In order to measure the impact of tools (*RRS*, *SCRS*) and their respective scales (*brooding*, *reflection* and *self-critical*) on the combined network structure, we constructed an *exponential random graph model* (ERGM; Hunter (2007)). In the model of our combined network predicting edges, *endogenous* predictors (related to the nature of the network structure) included the **number of edges** in the network (*edges* member), and a generalization of triadic structures based on alternating sums of *k-triangles* (Goodreau et al., 2009; Hunter, 2007) to involve some higher-order global network structures. This predictor served as a reflection on **clustering** in the form of completed triangles of vertices (we used *GWESP* (see Goodreau et al. (2009)), which stands for *geometrically weighted edgewise shared partner distribution*). *GWESP* has the advantage of taking a parametric form of the shared common vertex count distribution on each edge, giving each connecting vertex a declining positive probability of two selected vertices being connected. In order to capture *exogenous* effects (i.e., vertex attributes), we added *tool* and *sub-scale* as predictors, especially testing for *homophily* (assuming that belonging to the same tool and its scale will increase the log-odds of being connected by an edge).

2.5 Association Network Inference

In order to understand the nature of the edges of the combined association network, we utilized Association Network Inference techniques. The R package **fdrtool** (Korbinian Strimmer (2008), K. Strimmer (2008)) allowed us to model the probability density function and estimate η_0 of the fraction of true null hypotheses in the data; and $(1 - \eta_0)$ for the true alternative hypotheses. The mixture model for some statistic S thus takes the following form (where in our analysis, S is the partial correlation coefficient).

$$f(S) = \eta_0 f_0(S) + (1 - \eta_0) f_1(S)$$

To apply adjustment due to the number of performed tests, we controlled p-value by the *False Discovery Rate* (*FDR*; the rate of false rejections against the number of rejections).

2.6. Latent Network Modeling

We constructed an *eigenmodel* to further explore the structure of the combined graph of *RRS* and *SCRS* (Hoff et al. (2002); methodology following Mair (2018), Kolaczyk (2020)) and how well the network structure predicts edges. Our expectation was that sub-scales will have non-trivial impact on the resulting eigenvalues. In our two-part analysis, first we used eigenvalue decomposition in order to use the first two eigenvectors

reflective to our scale- and sub-scale structures of *RRS* and *SCRS*. In the second part, we defined three conditions in the *eigenmodel* on top of the original model. In the first iteration we used the original *eigenmodel* (discussed above) without the regression component, defining the adjacency matrix as function of only the pairwise latent effects summarized in the following matrix.

$$Y_{ij} = f(u_i \Lambda u_j)$$

In the second and third iterations, we defined an optional regression model in addition to the pairwise latent effects, where $\mathbf{x}_{ij}$ is a pair-specific predictor vector. This extends our model to the following equation.

$$Y_{ij} = f(\beta' x_{ij} + u_i \Lambda u_j)$$

Thus, in the second iteration we cross-validated an extended *eigenmodel* with the **tool** (*RRS* or *SCRS*) as nominal grouping factor of vertices playing important predictor role. And in the same manner, the third iteration tested a model with the scales of *brooding*, *reflection* and *self-critical* serving as pair-wise covariate predictor effects.

2.7. Power analysis, network accuracy and stability

Accuracy and stability have an important role in estimating network parameters when samples might vary in characteristics and size (Epskamp et al., 2017a, 2017b). One way of estimating network accuracy is to produce a 95% confidence interval of edge-weights with some variant of bootstrap. In the present paper, we chose the non-parametric bootstrap estimate for interpretability. Network stability is quantified with the *Correlation Stability* (*CS*-coefficient): an estimate of the maximum proportion of cases that can be dropped (with 95% certainty) to retain a correlation with the original centrality higher than some fixed threshold value (in the present paper, 0.7). Simulation studies (Epskamp et al., 2017a) lead us in our expectation as the *CS*-coefficient should be between 0.25 and 0.5.

3. Results

3.1 Network descriptives and community detection

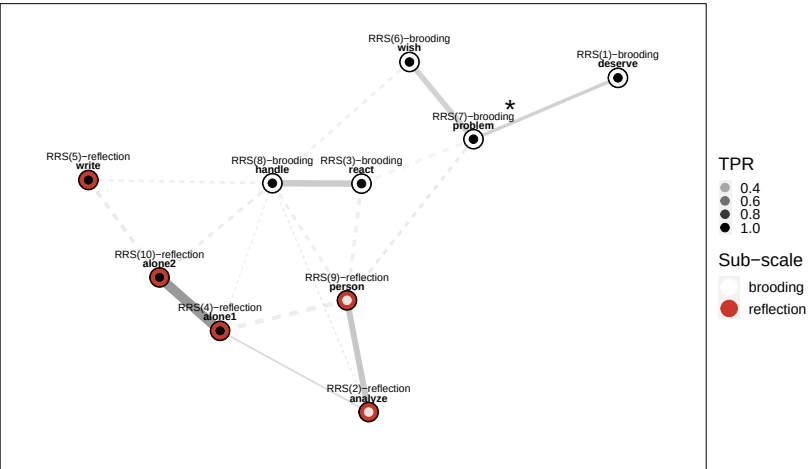
The undirected, weighted, associational networks of *RRS*, *SCRS* and their combined structure can be seen below (see Figure 1). All three networks proved to be connected (any vertex is reachable from every other). Comparing diameters to mean distances, we see densely connected graphs (for the *RRS* network, its diameter is 0.467 compared to its mean distance of 1.822; the *SCRS* and the combined networks show similar ratio patterns). On the plot, vertices represent the corresponding items of each questionnaire with labels showing item number, sub-scale and a label for each item. Sub-scales are presented in a colour-coded fashion as the outer circle of each vertex. Edges are the **LASSO** *L1*-penalized partial correlation coefficients whose value is represented as shade and width of the arcs.

The *RRS* (order of 10, size of 17) and *SCRS* (order of 10, size of 16) networks are densely connected networks. Analyzing their structure, while *SCRS* exhibits no *articulation point* (also called *cut vertex* whose removal would disconnect the network; Kolaczyk (2020)), the vertex of the ‘*problem*’ (“*What am I doing to deserve this?*”) item of *RRS* acts as an **articulation point**, as it would cut off the ‘*deserve*’ (“*Why do I have problems other people don’t have?*”) vertex from the rest of the network. Articulation points are marked with * on Figure 1. Besides **size** and **order**, on Figure 1. we also indicated the **Clustering coefficient** (*transitivity*), the **diameter** and the average vertex-to-vertex path lengths (**mean distance**).

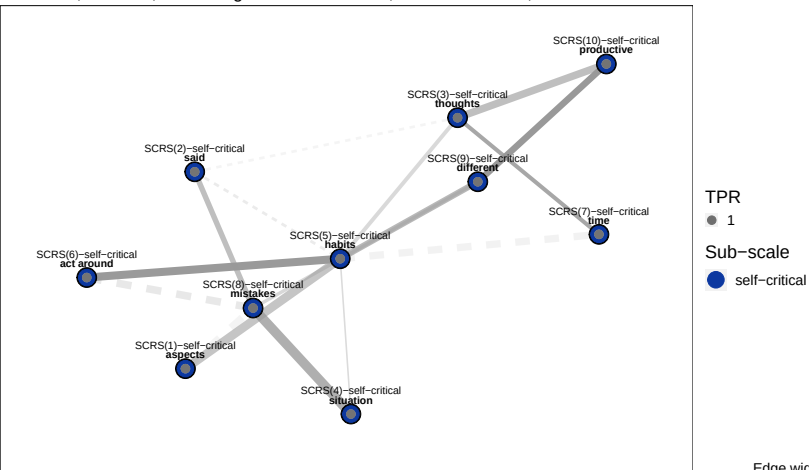
The combined network (order: 20, size: 39) has no articulation points. The previously identified articulation point of the *RRS* network (*problem* vertex) is no longer a cut-off vertex.

In order to analyze their respective internal structures, we performed community detection using the *igraph* package in R. On Figure 1., the inner node point of each vertex provides colour coding for our community detection results (the presented shades correspond to the True Positive Rates of each sub-scale; in other words, the average match between the detected community and the original sub-scale).

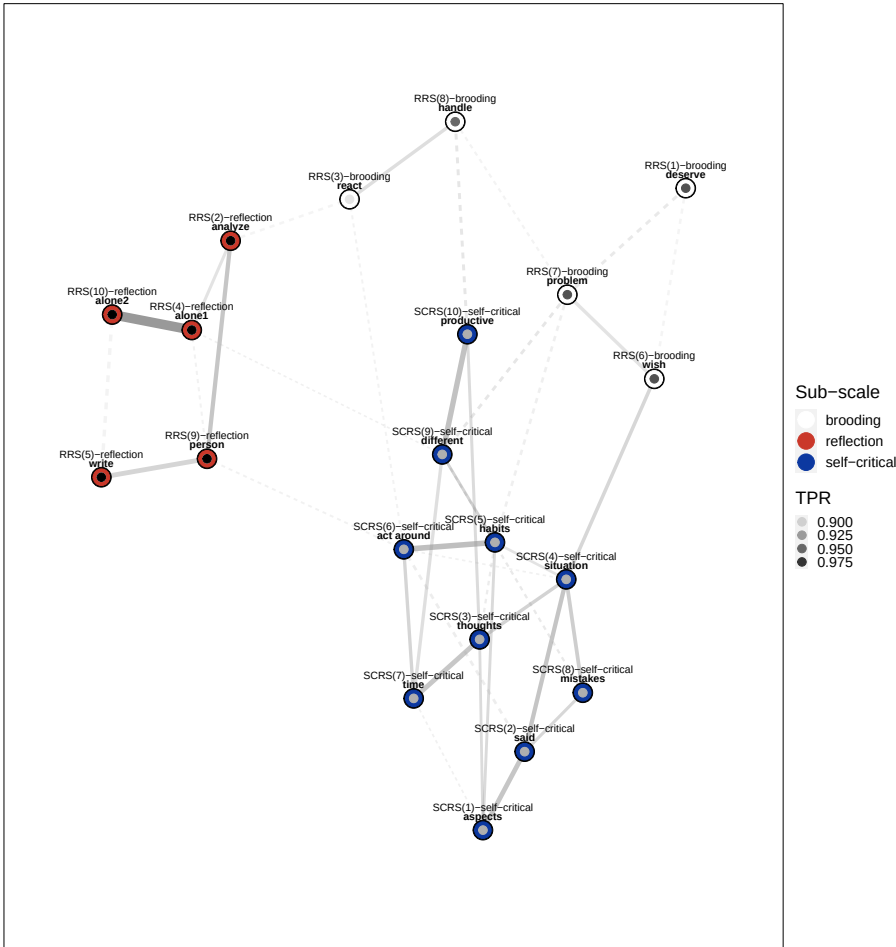
RRS undirected graph
Order=10; Size=17; Clustering Coefficient=0.444; Diameter=0.467; Mean distance=1.822



SCRS undirected graph
Order=10; Size=16; Clustering Coefficient=0.125; Diameter=0.868; Mean distance=1.733



RRS & SCRS undirected graph
Order=20; Size=39; Clustering Coefficient=0.206; Diameter=0.989; Mean distance=2.316



* marks Articulation Points (cut vertices; when such vertices are removed disconnect the graph)
TPR is average True Positive Rate with 1000 times reruns of community detection
Order is # of vertices; Size is # of edges
Edge width reflect edge weight (penalized part. corr.), while edge shade reflects lower bound of 95% CI of bootstrap accuracy estimate
Dashed edge line indicates the 95% CI of accuracy estimate ranges below 0.0

Figure 1: Tableau view of three graphs of rumination

As the choice of algorithm has non-trivial impact on the identified community structure, after having selected the “*optimal*”, the “*spinglass*”, the “*walktrap*” and (to also include hierarchical clustering) the “*fast greedy*” methods, we tested each a 1000 times on the three networks. We also provided an edge weight vector in the form of the *LASSO*-penalized partial correlation coefficients (or alternatively, omitted weights).

The **RRS** two-scale structure could be recreated with all four algorithms with the inclusion of the weight vector not having effect on the measures. For the sub-scale of *brooding* the average **true positive rate** (TPR) was 0.8; for *reflection* 0.743. In case of the *reflection* sub-scale, lower average TPR becomes tangible with the inspection of items 2 (“*analyze*”; “*Analyze recent events to try to understand why you are depressed*”) and 9 (“*person*”; “*Analyze your personality to try to understand why you are depressed*”) whose low TPR indicates that depending in the choice of community detection they have a chance of forming their own community or being classified as members of the *brooding* scale, rather than being part of *reflection*.

The one-scale structure of the *SCRS* has been reestablished.

The combined rumination network shows a clear expected structure, with having high TPR for each sub-scale (on average, *brooding* (0.9), *reflection* (1) and *self-critical* (0.8)). Commenting on the stability of the network community structure, the *RRS reflection* sub-scale exhibits strong TPR rates, and the weakness of the two items from their respective stand-alone network for items 2 and 9 are no longer present in the combined graph. The *RRS brooding* scale has good separation in terms of TPR, except for its item 3 (*react*) having <0.9 TPR (“*Why do I always react this way?*”). The one-scale *SCRS* has also a separable community with high (>0.925) TPR. The combined network has large TPR of 0.9; while keeping only the *spinglass* algorithm (as used by (Bernstein et al., 2019)), we get high average 1.0 TPR for the combined graph.

Please refer to our vertex summary table in the Appendix (Table 3.) for actual descriptives, including expected influence, bridge node related statistics and average True Positive Rates for each vertex. Please note where applicable, we used the standardized the scores.

3.2 Internal Structure with an exponential random graph model

Our ERGM model was built with *Markov chain Monte Carlo* (MCMC) maximum likelihood estimation (1000 iterations) to obtain model parameters by generating samples from the space of possible networks allowed by the data.

Table 1: ERGM model fit results with Markov chain Monte Carlo Maximum Likelihood Estimation

Term	Estimate	Std. Error	MCMC Error %	Statistic	p
# of edges	-2.238	0.605	0	-3.699	0.000
GWESP [triangle]	-0.373	0.161	0	-2.320	0.020
Tool (main effect)	-0.041	0.407	0	-0.100	0.920
Tool (homophily effect)	-0.825	1.240	0	-0.665	0.506
Scale (homophily effect)	4.185	1.336	0	3.133	0.002

An ANOVA analysis on the overall model indicates (with $df=5$, Residual Deviance=143.38, $p=0.0000$) that variables involved in the model explain the network connectivity in a highly non-trivial fashion. The detailed analysis reveals that *endogenous* effects such as the overall density of the network(number of *edges*) change the probability of the formation of an edge by 0.096), and at $p<0.05$ level the closed triangle formulation indicates a non-trivial impact on clustering (connectivity probability changed by 0.408). As expected, *exogenous* effects, such as two items belonging to the same sub-scale have a significant positive impact on connectivity (sub-scale changes edge probability by 0.985, keeping the rest of the network fixed). Tool in itself has no impact on edge formation. In terms of goodness-of-fit, observed frequencies of degree levels differ in non-significant manner from *MCMC*-simulated expectations. In two cases of vertex degree ($V_d=3$ and $V_d=6$), the observed frequencies are above the simulated average (6 > the expected avg. of 4.49 for $V_d=3$; and 4 > the expected avg. of 1.94 for $V_d=6$), but differences are not significant.

Regarding the expected edgewise shares partners, the only deviation from expected levels it for 1 partner, where the observed frequency is 17 against 13.28. Additional MCMC diagnostics revealed good fit of the model.

3.3 Association Network Inference

Results show low η_0 (0.051) indicates large mixing rate for the alternative hypothesis of edges having non-zero weights. The histogram of absolute correlation coefficients (see Figure 3. in Appendix) has a centered shape in contrast to the null component. The density-based *FDR* drops once empirical correlations reach $\rho=0.1$ (see the lowest section of Figure 3.).

3.4 Latent Network Modeling

Our next model was built with the *eigenmodel_mcmc* function of the R package **eigenmodel**. The function uses *Markov chain Monte Carlo* (MCMC) techniques to complete a model specification. By applying eigenvalue decomposition, we obtained the first two eigenvectors (see Figure 2).

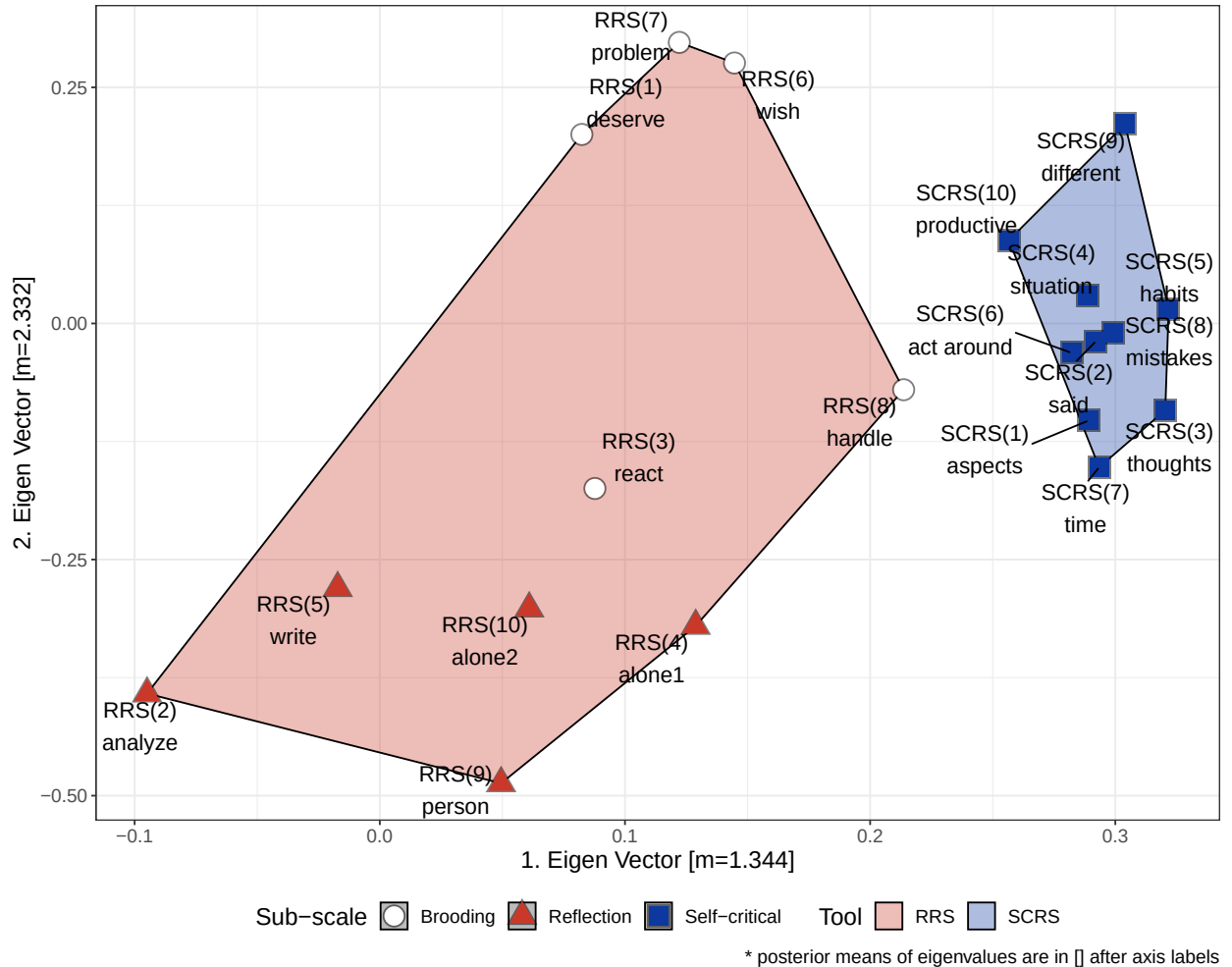


Figure 2: Vertices of the combined graph by the first two eigenvalues

Our visual insight from Figure 2. is that the first eigenvector serves as a separator between *RRS* and *SCRS* having a posterior mean of $e\bar{v}_1=1.344$. The second eigenvector seems to separate between the upper half of the two-dimensional space of *SCRS* and the *brooding* scale, and the lower half being the *reflection* scale ($e\bar{v}_2=2.332$).

In order to test the goodness-of-fit of the eigenmodel, using k -fold cross-validation ($k=5$), we refitted the model to a subset of the original data, using the complement 1/5 fraction for predicting the elements of the adjacency matrix of our original network. We used three different models in this part as discussed above (Section 2.6).

Summarizing the results from the *eigenmodels*, using the original combined network adjacency matrix as a reference, we estimated the *Area Under the Curve* (AUC) statistics for each model. Compared to the latent model ($AUC_{latent} = 0.348$), the inclusion of the tool-level grouping factor resulted in a better predictive model ($AUC_{latent+tool} = 0.637$). The final model involving the scale-level vertex characteristics has the best goodness-of-fit ($AUC_{latent+scale} = 0.725$). See Figure 4. in the Appendix for the corresponding ROC curves.

3.6 Power analysis, network accuracy and stability

We used the *bootnet* R package to have an estimate of edge-weight accuracy with *non-parametric bootstrap* (2500 times sampling). The resulting **accuracy** estimates have been visually incorporated in Figure 1 in the following manner. Edge shade reflects the lower bound of the 95% confidence interval (CI) for each edge, while edges are presented with dashed line for cases of the CI including 0.0 edge weights. For such cases, we cannot be sure if their weight persist given different samples and varying sample characteristics. Darker shaded, continuous edges are stable according to this bootstrap estimate and they preserve their sampled nature. Edges as between the *SCRS*(5) and *SCRS*(7) items indicate moderate penalized partial correlation between items in the SCRS network ($\rho=0.352$), but the edge is presented with dashed line and has lighter shade indicating a lower stability. Examples for accurate and strong edges are *RRS*(4) (*alone1*; “Go away by yourself and think about why you feel this way”) and *RRS*(10) (*alone2*; “Go someplace alone to think about your feelings”). The edge weight for these items in their stand-alone *RRS* network is 0.572 (estimated edge-weight accuracy is with average 0.511 with lower and upper CI bounds of 0.454 - 0.567). These items exhibit the same pattern in the combined network with estimated edge-weight accuracy with average weight of 0.493; lower and upper CI bounds of 0.436 - 0.549.

In the SCRS network we can track a *path* (a graph movement [*walk*] without repeated edges or vertices) between *SCRS* 6, 5, 9 and 10 items (please refer to Table 5. for edge-level weight and accuracy estimates of *SCRS*).

Such edges are less likely to exhibit similar characteristics with varying samples.

Network **stability** estimates for our three networks were carried out by using the *bootnet* function from the R package bearing the same name. We performed a *case dropping bootstrap* to simulate scenarios where fewer and fewer cases of the original sample would have been available and quantified the stability of the centrality metrics with the *Correlation Stability* (CS) coefficient. Bootstrap was performed with $B=2500$ resamples.

Accuracy and stability statistics have been summarized for *RRS*, *SCRS* and their combined graphs in respective tables of Appendix (see Table 4., 5. and 6).

As shown at (Epskamp et al., 2017a), the *Correlation Stability* coefficient is ideally between 0.25 and 0.5, the latter being the cut-off point for the estimate.

Table 2: Correlation Stability coefficient estimates for RRS, SCRS and their combined network

Metric	RRS	SCRS	RRS & SCRS
betweenness	0.00	0.05	0.00
closeness	0.13	0.28	0.00
edge	0.67	0.67	0.67
strength	0.21	0.28	0.52

Results indicate for betweenness centrality that neither networks tolerate much sample loss having CS-coefficients around 5% or lower. For the *RRS* and *SCRS* graphs, the closeness metrics are stable in the sense that around 13% and respectively 28% of the case can be dropped to still maintain 0.7 or higher correlation with their original estimates. The combined network’s closeness stability is low and rigid. Edge weights exhibit a constant stability of 67%. The edge strength *Correlation Stability* coefficients are 21% (RRS), 28% (SCRS) and 52% (combined graph) which indicates increased tolerance for the combined network of the two questionnaires.

4. Discussion

In the present paper, we examined the separation of two facets of rumination (a generic one, lying along the *brooding-reflection* axis; and the *self-critical* aspects of rumination) using a network approach. In its respective network the **Ruminative Response Style Questionnaire** exhibits a two-scale structure. Our community detection estimates suggest though that the reflective aspects of rumination regarding analyzing one’s self and strive to understand feelings (items 2 and 9 of *RRS*) have more in common with brooding, or have a tendency to emerge as a third aspect. The **Self-Critical Rumination Scale** proved to have a one-scale structure. Separation between scales in the combined graph is underlined by our eigenmodel, as in the two-dimensional space of the first two eigenvectors, expected grouping of *RRS* and *SCRS* items can be achieved. Items 2 and 9 are well separated from the bulk of the rest of the *RRS* items, supporting our above conclusion of them being a part of reflection or found a new component. Brooding has more associations with the self-critical aspects of rumination than reflection, which is highlighted by our eigenvector analysis as well as the sparse, unstable and weak edges with which reflection connects to brooding and self-critical rumination in their combined network.

We have gathered evidence of scale playing an important role in the edge-building process. According to our exponential random graph model, the internal scales of the combined network indeed have a non-trivial impact on associations between items, whereas the effect of questionnaire dissolved, no longer having an impact on the probability of edge formation. This suggests that all three scales are effective in assessing a separate facet of rumination when used together. As shown, associations within the combined network only have low chances of false discovery.

Results indicate that network associations are accurately aligned to the internal factor structure, while some associations are more sensitive to sampling characteristics and sampling variation. Edge strengths and edge weights have also high stability and relatively high resistance to sample loss, whereas betweenness is remarkably sensitive to case-dropping.

Future steps include analyzing such data from a person-level graph perspective, where vertices are individuals, and edges can be some distance measure between vertices based upon demographics and survey data. Also, if keeping our analysis on the item-level, as in the present paper, we would extend our methodology to directed acyclic graphs (DAGs) which would allow causal assumptions to support our understanding various facets of rumination.

Appendix

Table 3: Network descriptive statistics for RRS, SCRS and their combined networks

Title	Item	Scale	Label	Deg.	Str.	Bet.	Clo.	EI1	EI2	Br.Str.	Br.Bet.	Br.Cl.	Br.EI1	Br.EI2	TPR
RRS	RRS_1	brooding	deserve	1	0.14	0	0.33	-1.57	-1.60	-1.15	-0.89	-1.60	-1.15	-1.37	1.00
	RRS_2	reflection	analyze	3	0.34	0	0.78	-0.82	-0.70	-1.00	-0.89	-0.09	-1.00	-0.69	0.22
	RRS_3	brooding	react	3	0.65	0	0.41	0.28	0.40	0.46	-0.08	0.78	0.46	0.72	0.96
	RRS_4	reflection	alone1	4	0.82	0	0.72	0.91	1.17	0.22	0.32	0.71	0.22	0.24	1.00
	RRS_5	reflection	write	2	0.21	0	0.57	-1.29	-1.25	-0.76	-0.89	-1.43	-0.76	-1.02	0.99
	RRS_6	brooding	wish	2	0.38	0	0.51	-0.69	-0.70	-0.63	-0.89	-0.77	-0.63	-0.65	1.00
	RRS_7	brooding	problem	4	0.67	8	0.51	0.36	0.04	0.22	1.33	0.12	0.22	0.06	1.00
	RRS_8	brooding	handle	7	0.78	26	0.81	0.75	0.68	1.18	0.32	0.47	1.18	1.19	0.96
	RRS_9	reflection	person	5	0.88	16	0.68	1.11	0.95	1.94	1.94	1.55	1.94	1.77	0.22
	RRS_10	reflection	alone2	3	0.83	0	0.47	0.95	1.01	-0.49	-0.28	0.25	-0.49	-0.25	1.00
SCRS	SCRS_1	self-critical	aspects	2	0.77	0	0.19	-0.31	0.29	-0.16	0.53	0.84	-0.16	0.71	1.00
	SCRS_2	self-critical	said	3	0.59	3	0.30	-0.70	-0.75	-0.31	-0.74	-0.88	-0.31	-0.46	1.00
	SCRS_3	self-critical	thoughts	4	0.94	6	0.29	0.04	-0.27	0.51	-0.32	-0.44	0.51	0.06	1.00
	SCRS_4	self-critical	situation	2	0.56	0	0.27	-0.77	-0.73	-0.96	-0.74	-1.17	-0.96	-1.14	1.00
	SCRS_5	self-critical	habits	7	1.92	15	0.34	2.12	2.09	2.32	2.21	1.63	2.32	2.07	1.00
	SCRS_6	self-critical	act around	2	0.74	0	0.19	-0.37	0.16	-0.11	-0.74	0.96	-0.11	0.17	1.00
	SCRS_8	self-critical	mistakes	5	1.63	6	0.27	1.51	1.21	0.42	0.95	0.52	0.42	0.22	1.00
	SCRS_9	self-critical	different	3	0.80	3	0.25	-0.25	-0.11	0.22	0.32	0.37	0.22	0.42	1.00
	SCRS_7	self-critical	time	2	0.57	0	0.21	-0.75	-0.82	-0.63	-0.74	-0.50	-0.63	-0.60	1.00
	SCRS_10	self-critical	productive	2	0.67	0	0.19	-0.52	-1.09	-1.30	-0.74	-1.32	-1.30	-1.44	1.00
Combined	RRS_1	brooding	deserve	2	0.31	0	0.11	-1.82	-1.80	-1.00	-1.05	-0.91	-1.00	-0.91	0.96
	RRS_2	reflection	analyze	3	0.57	2	0.13	-0.74	-0.79	-0.17	-0.35	-0.86	-0.17	-0.48	1.00
	RRS_3	brooding	react	3	0.45	15	0.15	-1.23	-1.33	0.47	-0.08	0.23	0.47	0.23	0.90
	RRS_4	reflection	alone1	4	0.94	23	0.17	0.88	0.71	-0.51	0.45	-1.15	-0.51	-0.57	1.00
	RRS_5	reflection	write	2	0.48	11	0.09	-1.12	-1.11	-1.00	-0.88	-1.45	-1.00	-1.16	1.00
	RRS_6	brooding	wish	3	0.55	3	0.12	-0.81	-0.78	0.18	-0.52	0.55	0.18	0.37	0.96
	RRS_7	brooding	problem	5	0.80	26	0.16	0.28	0.06	1.50	0.81	1.11	1.50	1.49	0.96
	RRS_8	brooding	handle	3	0.49	7	0.14	-1.08	-1.23	-0.31	-0.35	-0.20	-0.31	-0.37	0.95
	RRS_9	reflection	person	4	0.75	31	0.16	0.05	-0.21	-0.38	1.07	-0.66	-0.38	-0.41	1.00
	SCRS_1	self-critical	aspects	4	0.69	5	0.14	-0.19	-0.01	-1.00	-1.05	-0.83	-1.00	-0.94	0.92
	SCRS_2	self-critical	said	4	0.86	0	0.13	0.51	0.53	-1.00	-0.70	-0.58	-1.00	-0.90	0.92
	SCRS_3	self-critical	thoughts	5	0.94	6	0.14	0.87	0.90	-0.01	-0.52	0.34	-0.01	0.07	0.92
	SCRS_4	self-critical	situation	6	1.11	28	0.18	1.61	1.47	0.18	0.19	0.31	0.18	0.12	0.92

Title	Item	Scale	Label	Deg.	Str.	Bet.	Clo.	EI1	EI2	Br.Str.	Br.Bet.	Br.Cl.	Br.EI1	Br.EI2	TPR
	SCRS_5	self-critical	habits	6	1.08	12	0.17	1.47	1.70	0.80	1.34	1.26	0.80	0.89	0.92
	SCRS_6	self-critical	act around	6	0.99	51	0.18	1.06	1.15	0.26	2.22	0.83	0.26	0.39	0.92
	SCRS_7	self-critical	time	4	0.76	9	0.14	0.11	0.34	0.10	-0.52	0.72	0.10	0.07	0.92
	SCRS_8	self-critical	mistakes	3	0.52	2	0.14	-0.94	-0.65	-0.27	-0.70	-0.37	-0.27	-0.43	0.92
	SCRS_9	self-critical	different	6	1.05	38	0.18	1.32	1.16	3.19	2.04	2.18	3.19	3.02	0.92
	RRS_10	reflection	alone2	2	0.76	0	0.07	0.11	0.13	-1.00	-0.88	-1.48	-1.00	-1.10	1.00
	SCRS_10	self-critical	productive	3	0.66	1	0.12	-0.35	-0.23	-0.01	-0.52	0.93	-0.01	0.63	0.92

Deg.=Degree; Str.=Strength; Bet.=Betweenness; Clo.=Closeness; EI1= One-step Expected Influence; EI2= Two-step Expected Influence; Br.Str.=Bridge Strength; Br.Bet.=Bridge Betweenness; Br.Cl.=Bridge Closeness; Br.EI1=One-step Bridge Expected Influence; Br.EI2=Two-step Bridge Expected Influence; TPR=average True Positive Rate

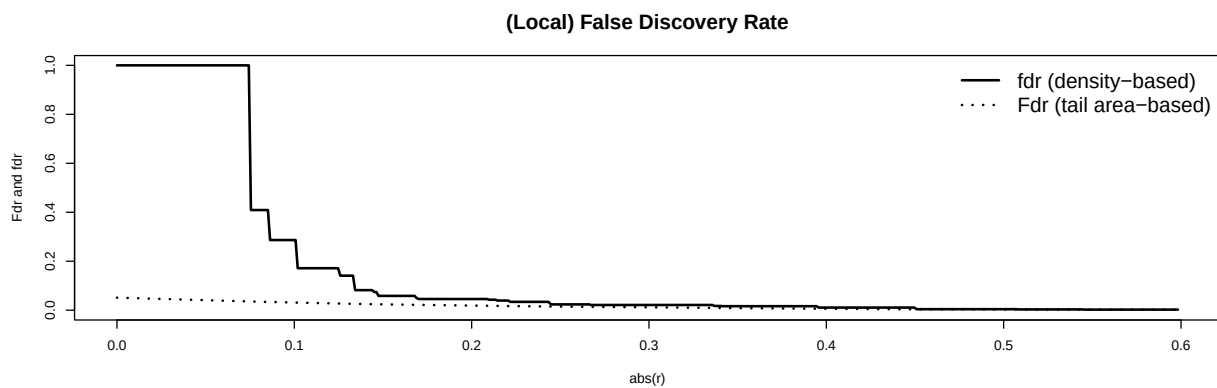
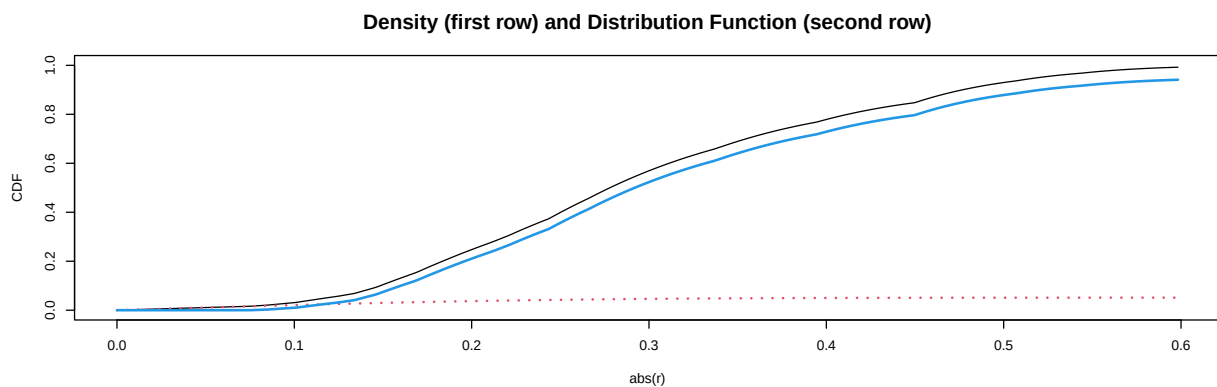
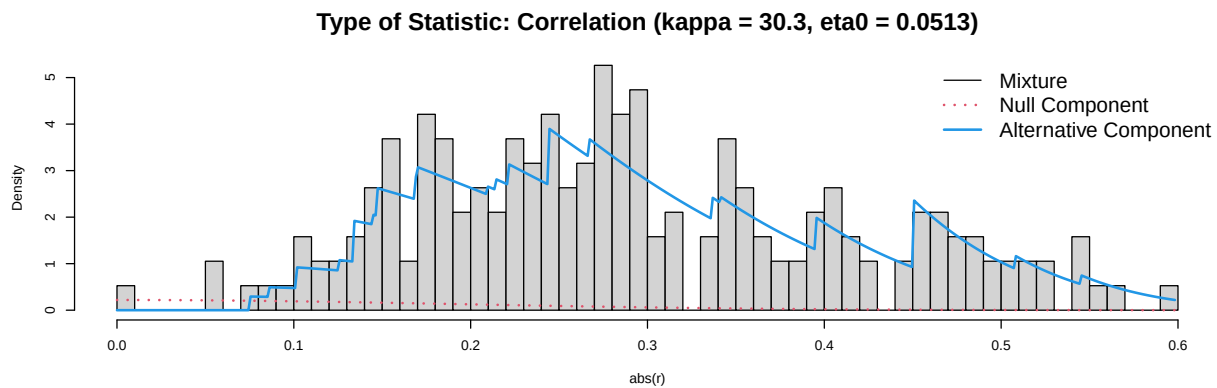


Figure 3: False Discovery Rate (FDR) for the combined network

Table 4: Edge Accuracy and Stability Statistics for the RRS network

Vx.1	Vx.2	Avg. Acc.	L.B.Acc.	U.B.Acc.	Miss.:%:0	Miss.:%:0.2	Miss.:%:0.4	Miss.:%:0.6	Miss.:%:0.7
RRS_1	RRS_7	0.20	0.13	0.27	0.20	0.20	0.19	0.18	0.12
RRS_2	RRS_4	0.15	0.08	0.22	0.16	0.15	0.14	0.12	0.08
RRS_2	RRS_8	0.00	-0.01	0.01	0.00	0.00	0.00	0.00	0.00
RRS_2	RRS_9	0.25	0.19	0.32	0.25	0.25	0.26	0.24	0.22
RRS_3	RRS_7	0.06	-0.07	0.18	0.01	0.01	0.02	0.02	0.02
RRS_3	RRS_8	0.23	0.16	0.30	0.23	0.23	0.23	0.22	0.17
RRS_3	RRS_9	0.09	-0.04	0.22	0.10	0.07	0.05	0.04	0.02
RRS_4	RRS_10	0.51	0.45	0.57	0.51	0.51	0.51	0.51	0.50
RRS_4	RRS_8	0.06	-0.05	0.17	0.06	0.04	0.02	0.02	0.02
RRS_4	RRS_9	0.09	-0.03	0.20	0.10	0.08	0.05	0.05	0.03
RRS_5	RRS_10	0.11	0.00	0.22	0.12	0.11	0.07	0.06	0.03
RRS_5	RRS_8	0.00	-0.04	0.04	0.00	0.00	0.00	0.00	0.00
RRS_6	RRS_7	0.20	0.13	0.27	0.20	0.20	0.19	0.17	0.10
RRS_6	RRS_8	0.09	-0.04	0.22	0.11	0.07	0.05	0.03	0.02
RRS_7	RRS_9	0.00	-0.01	0.01	0.00	0.00	0.00	0.00	0.00
RRS_8	RRS_10	0.00	-0.03	0.04	0.00	0.00	0.00	0.00	0.00
RRS_8	RRS_9	0.09	-0.03	0.22	0.11	0.09	0.05	0.04	0.03

Vx.1=1st Vertex the edge attaches to; Vx.2=2nd Vertex the edge attaches to; Avg.Acc.=Average Edge Weight by Bootstrap; L.B.Acc.=Lower Bound of 95% CI for bootstrapped network accuracy; U.P.Acc.=Upper Bound of 95% CI for bootstrapped network accuracy; Statistic=type of network characteristic; Miss.%=Estimated average stability at the given missing rate of sample size

Table 5: Edge Accuracy and Stability Statistics for the SCRS network

Vx.1	Vx.2	Avg. Acc.	L.B.Acc.	U.B.Acc.	Miss.:%:0	Miss.:%:0.2	Miss.:%:0.4	Miss.:%:0.6	Miss.:%:0.7
SCRS_1	SCRS_5	0.16	0.09	0.23	0.16	0.16	0.16	0.15	0.13
SCRS_1	SCRS_8	0.04	-0.06	0.13	0.00	0.02	0.02	0.01	0.02
SCRS_2	SCRS_3	0.03	-0.06	0.11	0.00	0.01	0.02	0.01	0.02
SCRS_2	SCRS_5	0.00	-0.03	0.03	0.00	0.00	0.00	0.00	0.00
SCRS_2	SCRS_8	0.18	0.11	0.25	0.18	0.18	0.18	0.17	0.15
SCRS_3	SCRS_10	0.18	0.11	0.24	0.18	0.18	0.18	0.17	0.14
SCRS_3	SCRS_5	0.11	0.03	0.19	0.11	0.11	0.10	0.09	0.07
SCRS_3	SCRS_7	0.25	0.18	0.32	0.25	0.25	0.25	0.25	0.25
SCRS_4	SCRS_5	0.11	0.02	0.20	0.12	0.11	0.10	0.09	0.07
SCRS_4	SCRS_8	0.23	0.16	0.29	0.23	0.23	0.23	0.23	0.22
SCRS_5	SCRS_6	0.29	0.23	0.36	0.30	0.30	0.29	0.30	0.29
SCRS_5	SCRS_7	0.06	-0.05	0.16	0.04	0.04	0.04	0.04	0.04
SCRS_5	SCRS_9	0.20	0.13	0.27	0.20	0.20	0.20	0.19	0.19
SCRS_6	SCRS_8	0.09	-0.02	0.19	0.10	0.08	0.06	0.05	0.05
SCRS_8	SCRS_9	0.12	0.03	0.20	0.12	0.12	0.11	0.08	0.05
SCRS_9	SCRS_10	0.30	0.23	0.36	0.30	0.30	0.30	0.29	0.29

Vx.1=1st Vertex the edge attaches to; Vx.2=2nd Vertex the edge attaches to; Avg.Acc.=Average Edge Weight by Bootstrap; L.B.Acc.=Lower Bound of 95% CI for bootstrapped network accuracy; U.P.Acc.=Upper Bound of 95% CI for bootstrapped network accuracy; Statistic=type of network characteristic; Miss.%=Estimated average stability at the given missing rate of sample size

Table 6: Edge Accuracy and Stability Statistics for the RRS and SCRS network

Vx.1	Vx.2	Avg. Acc.	L.B.Acc.	U.B.Acc.	Miss.:%:0	Miss.:%:0.2	Miss.:%:0.3	Miss.:%:0.6	Miss.:%:0.7
RRS_1	RRS_6	0.06	-0.09	0.22	0.00	0.01	0.01	0.00	0.01
RRS_1	RRS_7	0.13	-0.03	0.29	0.16	0.11	0.07	0.03	0.01
RRS_2	RRS_3	0.06	-0.09	0.21	0.00	0.01	0.01	0.01	0.01
RRS_2	RRS_4	0.13	0.01	0.26	0.15	0.13	0.10	0.04	0.02
RRS_2	RRS_9	0.24	0.17	0.30	0.24	0.24	0.24	0.19	0.11
RRS_3	RRS_8	0.16	0.04	0.28	0.17	0.16	0.13	0.06	0.02
RRS_3	SCRS_6	0.03	-0.08	0.14	0.00	0.00	0.00	0.00	0.00
RRS_4	RRS_10	0.49	0.44	0.55	0.49	0.49	0.49	0.49	0.46
RRS_4	RRS_9	0.06	-0.08	0.19	0.01	0.02	0.01	0.02	0.01
RRS_4	SCRS_9	0.02	-0.07	0.10	0.00	0.00	0.00	0.00	0.00
RRS_5	RRS_10	0.06	-0.08	0.21	0.01	0.01	0.01	0.00	0.00
RRS_5	RRS_9	0.19	0.09	0.29	0.20	0.19	0.17	0.08	0.03
RRS_6	RRS_7	0.14	0.00	0.28	0.16	0.13	0.09	0.04	0.02
RRS_6	SCRS_4	0.17	0.08	0.26	0.17	0.17	0.15	0.09	0.05
RRS_7	RRS_8	0.06	-0.08	0.21	0.00	0.01	0.01	0.01	0.00
RRS_7	SCRS_5	0.07	-0.06	0.21	0.03	0.03	0.03	0.02	0.01
RRS_7	SCRS_9	0.12	-0.01	0.25	0.14	0.11	0.07	0.05	0.04
RRS_8	SCRS_10	0.12	-0.01	0.25	0.15	0.12	0.09	0.04	0.03
RRS_9	SCRS_6	0.02	-0.07	0.12	0.00	0.00	0.00	0.00	0.00
SCRS_1	SCRS_2	0.24	0.17	0.30	0.24	0.24	0.24	0.23	0.18
SCRS_1	SCRS_3	0.14	0.05	0.23	0.15	0.15	0.13	0.09	0.05
SCRS_1	SCRS_5	0.15	0.06	0.24	0.16	0.15	0.14	0.11	0.06
SCRS_1	SCRS_7	0.05	-0.07	0.17	0.01	0.01	0.02	0.01	0.01
SCRS_2	SCRS_4	0.24	0.18	0.31	0.24	0.24	0.24	0.24	0.19
SCRS_2	SCRS_6	0.07	-0.05	0.19	0.07	0.04	0.03	0.01	0.02
SCRS_2	SCRS_8	0.16	0.08	0.24	0.16	0.16	0.15	0.11	0.07
SCRS_3	SCRS_10	0.15	0.05	0.24	0.16	0.16	0.14	0.10	0.06
SCRS_3	SCRS_4	0.17	0.10	0.25	0.17	0.17	0.17	0.14	0.09
SCRS_3	SCRS_5	0.08	-0.04	0.20	0.09	0.06	0.05	0.03	0.02
SCRS_3	SCRS_7	0.23	0.17	0.30	0.24	0.24	0.23	0.23	0.16
SCRS_4	SCRS_5	0.11	0.00	0.22	0.12	0.11	0.09	0.06	0.04
SCRS_4	SCRS_6	0.03	-0.07	0.14	0.00	0.01	0.01	0.01	0.01
SCRS_4	SCRS_8	0.20	0.13	0.27	0.20	0.20	0.20	0.18	0.13
SCRS_5	SCRS_6	0.29	0.22	0.36	0.29	0.29	0.29	0.28	0.25
SCRS_5	SCRS_9	0.17	0.10	0.25	0.17	0.17	0.17	0.15	0.10

Vx.1	Vx.2	Avg. Acc.	L.B.Acc.	U.B.Acc.	Miss.:%:0	Miss.:%:0.2	Miss.:%:0.3	Miss.:%:0.6	Miss.:%:0.7
SCRS_6	SCRS_7	0.17	0.09	0.25	0.17	0.17	0.17	0.12	0.08
SCRS_7	SCRS_9	0.14	0.03	0.24	0.14	0.14	0.12	0.09	0.06
SCRS_8	SCRS_9	0.09	-0.03	0.21	0.12	0.09	0.06	0.03	0.02
SCRS_9	SCRS_10	0.26	0.20	0.33	0.27	0.27	0.27	0.27	0.21

Vx.1=1st Vertex the edge attaches to; Vx.2=2nd Vertex the edge attaches to; Avg.Acc.=Average Edge Weight by Bootstrap; L.B.Acc.=Lower Bound of 95% CI for bootstrapped network accuracy; U.P.Acc.=Upper Bound of 95% CI for bootstrapped network accuracy; Statistic=type of network characteristic; Miss.%=Estimated average stability at the given missing rate of sample size

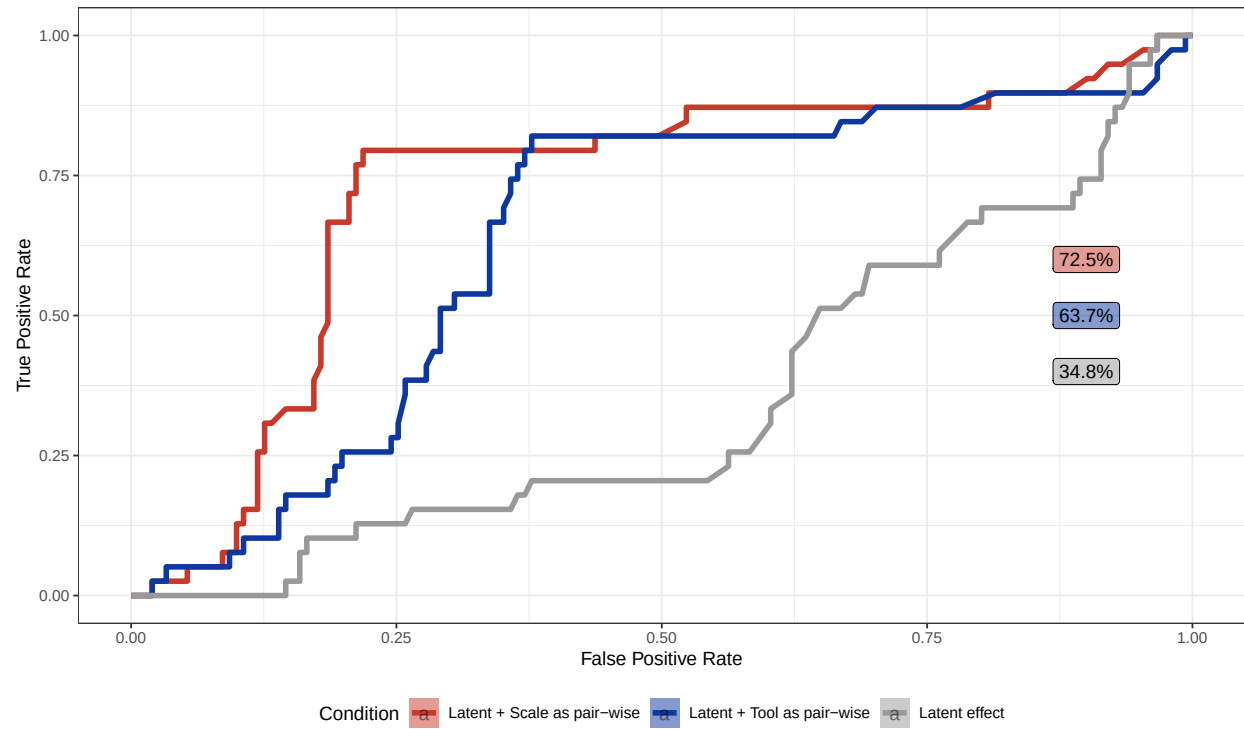


Figure 4: ROC curves of the combined RRS-SCRS network

References

- Bernstein, E. E., Heeren, A., & McNally, R. J. (2019). Reexamining trait rumination as a system of repetitive negative thoughts: A network analysis. *Journal of Behavior Therapy and Experimental Psychiatry*, 63, 21–27. <https://doi.org/10.1016/j.jbtep.2018.12.005>
- Donald J. Robinaugh, R. J. M., Alexander J. Millner. (2016). Identifying highly influential nodes in the complicated grief network. *J Abnorm Psychol.*, 125(6). <https://doi.org/10.1037/abn0000181>
- Epskamp, S., Borsboom, D., & Fried, E. I. (2017a). Estimating psychological networks and their accuracy: A tutorial paper. *Behavior Research Methods*, 50(1), 195–212. <https://doi.org/10.3758/s13428-017-0862-1>
- Epskamp, S., Borsboom, D., & Fried, E. I. (2017b). *Estimating psychological networks and their accuracy: Supplementary materials*.
- Epskamp, S., Cramer, A. O. J., Waldorp, L. J., Schmittmann, V. D., & Borsboom, D. (2012). Qgraph: Network visualizations of relationships in psychometric data. *Journal of Statistical Software*, 48(4). <https://doi.org/10.18637/jss.v048.i04>
- Epskamp, S., & Fried, E. I. (2018). A tutorial on regularized partial correlation networks. *Psychological Methods*, 23(4), 617–634. <https://doi.org/10.1037/met0000167>
- Friedman, J., Hastie, T., & Tibshirani, R. (2007). Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9(3), 432–441. <https://doi.org/10.1093/biostatistics/kxm045>
- Goodreau, S. M., Kitts, J. A., & Morris, M. (2009). Birds of a feather, or friend of a friend? Using exponential random graph models to investigate adolescent social networks. *Demography*, 46(1), 103–125. <https://doi.org/10.1353/dem.0.0045>
- Heeren, A., Jones, P. J., & McNally, R. J. (2018). Mapping network connectivity among symptoms of social anxiety and comorbid depression in people with social anxiety disorder. *Journal of Affective Disorders*, 228, 75–82. <https://doi.org/10.1016/j.jad.2017.12.003>
- Hoff, P. D., Raftery, A. E., & Handcock, M. S. (2002). Latent space approaches to social network analysis. *Journal of the American Statistical Association*, 97(460), 1090–1098. <https://doi.org/10.1198/016214502388618906>
- Hunter, D. R. (2007). Curved exponential family models for social networks. *Social Networks*, 29(2), 216–230. <https://doi.org/10.1016/j.socnet.2006.08.005>
- Jones, P. (2017). *Networktools: Assorted and tools for identifying and important nodes and in networks*.
- Kolaczyk, G. C. E. D. (2020). *Statistical analysis of network data with r*. Springer-Verlag GmbH. https://www.ebook.de/de/product/39334469/gabor_csardi_eric_d_kolaczyk_statistical_analysis_of_network_data_with_r.html
- Mair, P. (2018). *Modern psychometrics with r*. Springer International Publishing. https://www.ebook.de/de/product/33047896/patrick_mair_modern_psychometrics_with_r.html
- McNally, R. J., Mair, P., Mugno, B. L., & Riemann, B. C. (2017). Co-morbid obsessive-compulsive disorder and depression: A bayesian network approach. *Psychological Medicine*, 47(7), 1204–1214. <https://doi.org/https://doi.org/10.1017/S0033291716003287>
- Newman, M. E. J. (2010). *Networks - an introduction*. Oxford University Press.
- Reichardt, J., & Bornholdt, S. (2006). Statistical mechanics of community detection. *Physical Review*, 74(1).
- Strimmer, K. (2008). Fdrtool: A versatile r package for estimating local and tail area-based false discovery rates. *Bioinformatics*, 24(12), 1461–1462. <https://doi.org/10.1093/bioinformatics/btn209>
- Strimmer, Korbinian. (2008). A unified approach to false discovery rate estimation. *BMC Bioinformatics*, 9(1). <https://doi.org/10.1186/1471-2105-9-303>
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1), 267–288. <https://doi.org/10.1111/j.2517-6161.1996.tb02080.x>